

**ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)**  
**ORGANISATION OF ISLAMIC COOPERATION (OIC)**  
**Department of Computer Science and Engineering (CSE)**

QUIZ-1

DURATION: 25 MINUTES

SUMMER SEMESTER, 2023-2024

FULL MARKS: 20

## CSE 4411: Data Communication & Networking

Programmable calculators are not allowed.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

STUDENT ID:

1. Suppose users share a 3 Mbps link. Also, suppose each user requires 1 Mbps when transmitting, but each user transmits only 20 percent of the time.

Given,

 $X \equiv$  Number of devices that are active at the same time

$$P(X = k) = {}^n C_k (p)^k (1 - p)^{n-k}$$

- a) When circuit switching is used, how many users can be supported?

1  
(CO1)  
(PO1)

**Solution:**3 users,  $3 \text{ Mbps} / 1 \text{ Mbps} = 3$ **Rubric:**

- 1 point for correct answer.

- b) Suppose packet switching is used. Find the probability that a given user is transmitting

1  
(CO1)  
(PO1)

**Solution:**

.2, as a given user is active only 20 percent of the time

**Rubric:**

- 1 point for the correct answer.

- c) In case of packet switching, what is the minimum number of users that are needed to create any congestion? What is the probability of having any sort of congestion with the minimum users needed for congestion? In case of 125 users, what would be the probability of having any sort of congestion? Show all the calculations

5  
(CO1)  
(PO1)

**Solution:****Question 1:**

The link can support up to

$$\frac{3 \text{ Mbps}}{1 \text{ Mbps/user}} = 3$$

simultaneous users without congestion. Hence to *allow* more than 3 users to be active simultaneously, we require

$$N_{\min} = 4.$$

**Question 2:**

Let  $X$  be the number of active users at an instant. Then

$$X \sim \text{Binomial}(N = 4, p = 0.2).$$

Congestion occurs if  $X \geq 4$ . Thus

$$P(\text{congestion}) = P(X \geq 4) = P(X = 4) = \binom{4}{4} (0.2)^4 (0.8)^0 = (0.2)^4 = 0.0016.$$

$$P(\text{congestion} \mid N = 4) = 0.0016 \quad (0.16\%)$$

**Question 3:**

Let  $X$  be the number of active users at an instant. Then

$$X \sim \text{Binomial}(N = 125, p = 0.2).$$

Congestion occurs if  $X \geq 4$ . Thus

$$P(\text{congestion}) = 1 - P(X \leq 3) = 1 - \sum_{k=0}^3 \binom{125}{k} (0.2)^k (0.8)^{125-k}.$$

Numerically,

$$P(\text{congestion}) \approx 1 - [P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)] \approx 0.999999996.$$

$$P(\text{congestion} \mid N = 125) \approx 0.999999996 \quad (99.9999996\%)$$

**Rubric**

- 1 points for identifying the minimum number of users for congestion
- 2 points for each correct calculation

2. Answer true or false with one line of justification

$2 * 3 = 6$

- a) A user requests a Web page that consists of some text and three images. For this page, the client will send one request message and receive four response messages. (CO2) (PO2)

**Solution:**

False.

There are four connections since each connection transports exactly one request message and one response message. So, each object will have its one request message instead of there being only one request message.

- b) Two distinct Web pages (for example, [www.iut-oic.edu/research.html](http://www.iut-oic.edu/research.html) and [www.iut-oic.edu/s-tudents.html](http://www.iut-oic.edu/s-tudents.html)) can be sent over the same persistent connection

**Solution:**

True.

It is because both of these web pages are on the same physical server ([www.iut-oic.edu](http://www.iut-oic.edu)).

- c) With nonpersistent connections between browser and origin server, it is possible for a single TCP segment to carry two distinct HTTP request messages.

**Solution:**

False.

In a nonpersistent connection, the connection closes after each connection. In this case, the connection will close once the first message is received, and there will be a new connection opened to send the second message.

3. Suppose you and a friend are texting via WhatsApp, but for a while your messages aren't being delivered. You have confirmed your Internet connection by loading other websites, and everything else works fine. Your friend then calls you, and you learn that his WhatsApp is doing fine.

- a) Which of the five network layers might be responsible for this issue? Justify your answer, and explain why the other layers aren't relevant.

5  
(CO1)  
(PO2)

**Solution:**

Layers that may be responsible:

1. Application layer: Implements the application-specific protocol logic. The underlying transport may work, but the app's protocol exchanges never complete.
2. Transport layer: Provides end-to-end data delivery. If a transport session can't establish or repeatedly times out/reset, only that application's data stalls—even though lower layers still carry other flows successfully.

Why other layers are not relevant:

1. Network layer: Handles routing and addressing (IP). If IP were broken, you wouldn't reach any remote host—yet you can.
2. Datalink layer: Handles framing, error detection, and local retransmission. A DL failure blocks all IP traffic on that link.
3. Physical layer: Deals with raw bit transmission over a medium. If it failed, all services would stop, not just one.

**Rubric:**

- .5 point for each correct layer name
- .5 point for each correct justification

- b) What if your friend's WhatsApp is also not working fine? Will the same answer hold? If not justify your new answer

2  
(CO1)  
(PO2)

**Solution:**

Yes, Same

**Rubric:**

- 2 point for the correct answer